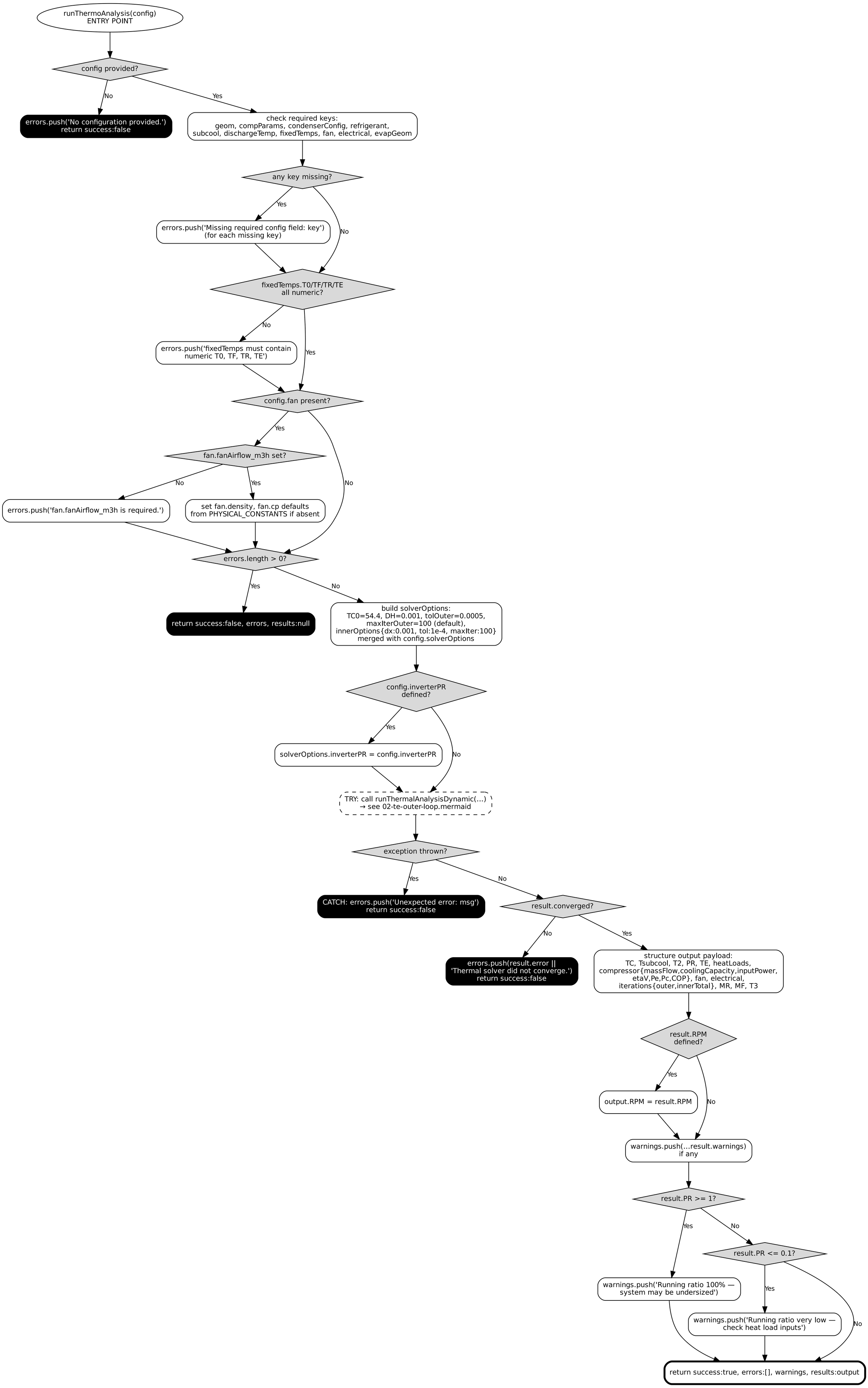
runThermoAnalysis(config)
ENTRY POINT
config provided?
No
Yes
errors.push('No configuration provided.')
return success:false
check required keys:
geom, compParams, condenserConfig, refrigerant,
subcool, dischargeTemp, fixedTemps, fan, electrical, evapGeom
any key missing?
Yes
No
errors.push('Missing required config field: key')
(for each missing key)
fixedTemps.T0/TF/TR/TE
all numeric?
No
Yes
errors.push('fixedTemps must contain
numeric T0, TF, TR, TE')
config.fan present?
Yes
No
fan.fanAirflow_m3h set?
No
Yes
errors.push('fan.fanAirflow_m3h is required.')
set fan.density, fan.cp defaults
from PHYSICAL_CONSTANTS if absent
errors.length > 0?
Yes
No
return success:false, errors, results:null
build solverOptions:
TC0=54.4, DH=0.001, tolOuter=0.0005,
maxIterOuter=100 (default),
innerOptions{dx:0.001, tol:1e-4, maxIter:100}
merged with config.solverOptions
config.inverterPR
defined?
Yes
No
solverOptions.inverterPR = config.inverterPR
TRY: call runThermalAnalysisDynamic(...)
→ see 02-te-outer-loop.mermaid
exception thrown?
Yes
No
CATCH: errors.push('Unexpected error: msg')
return success:false
result.converged?
No
Yes
errors.push(result.error ||
'Thermal solver did not converge.')
return success:false
structure output payload:
TC, Tsubcool, T2, PR, TE, heatLoads,
compressor{massFlow,coolingCapacity,inputPower,
etaV,Pe,Pc,COP}, fan, electrical,
iterations{outer,innerTotal}, MR, MF, T3
result.RPM
defined?
Yes
No
output.RPM = result.RPM
warnings.push(...result.warnings)
if any
result.PR >= 1?
Yes
No
warnings.push('Running ratio 100% —
system may be undersized')
result.PR <= 0.1?
Yes
No
warnings.push('Running ratio very low —
check heat load inputs')
return success:true, errors:[], warnings, results:output